Half-Wave Dipole Antenna

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Lecture Objective

By the end of this lecture, you will be able to:

- Evaluate the radiation properties of the half-wave dipole.

Wire and Metal Tube Antennas

- Wire antennas are simple and easy to construct as well as reasonably robust.
- Metal (commonly aluminum) tubing can also be used to construct antennas.
- We usually understand the category of “wire antennas” to include those antennas constructed with metal tubing.
- The half-wave dipole is a very popular antenna realized using wire or metal tubing.

Source of Radiation for Wire Antennas

- The source of the radiation emanating from a wire antenna can be modeled as the current distribution along the length of the wire or tube.
- The fields arising from the current distribution is obtained as a weighted superposition of the fields arising from Hertzian dipoles.

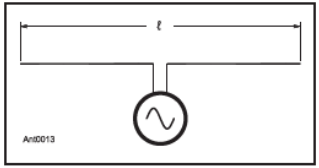

Figure 2.1 — The center-fed dipole antenna. It is assumed that the source of power is directly at the antenna feed point, with no intervening transmission line. Although λ/2 is the most common length for amateur dipoles, the length of a dipole antenna can be any fraction of a wavelength.

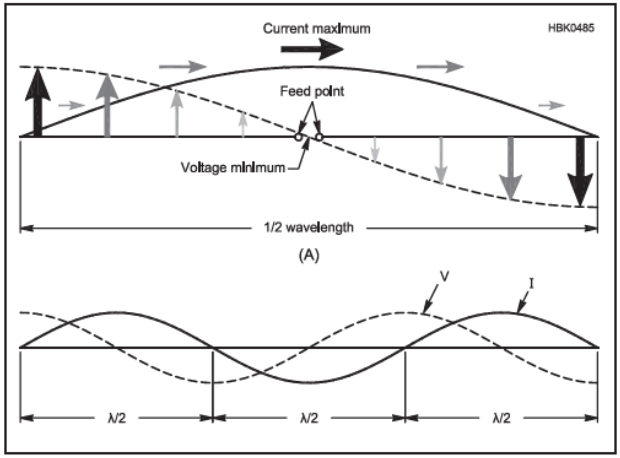

Figure 2.2 — The current and voltage distribution along a half-wave dipole (A) and for an antenna made from a series of half-wave sections (B).

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Overview of Half-Wave Dipole

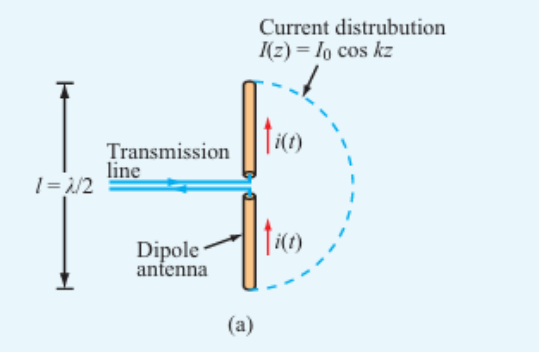
- The half-wave dipole (and the closely related quarter-wave monopole with large ground plane) is an extremely popular antenna.
- One reason for its popularity is that its input impedance is purely (or almost purely) real and close to 50 ohms, so (at least theoretically) an isolated half-wave dipole is well matched to a 50-ohm coaxial cable.
- The actual feed point impedance seen in practice for a specific half-wave dipole depends on other factors including height above ground, conductivity of ground, proximity to structures and natural features, etc.

Assumed Current Distribution Approach

- To accurately determine the current on a given wire dipole, we can use a *potentially-exact* numerical method; most commonly we use the Method of Moments to solve an *integral equation*.
- These numerical methods require 'discretization' of the problem into a finite set of unknown coefficients. A potentially-exact method is one where the numerical solution converges to the exact solution as the number of unknowns increases.
- Alternately, we can use physical insight to assume the current distribution. Although less accurate than properly applied numerical methods, the assumed current distribution approach is faster and provides more physical insight.

Assumed Phasor Current Distribution

- The phasor current distribution on the half-wave dipole is <u>assumed</u> to be



$$\tilde{I}(z) = I_0 \cos k_0 z, \ -l/2 \le z \le l/2$$

$$= I_0 \cos\left(2\pi \frac{z}{\lambda}\right), \ -\lambda/4 \le z \le \lambda/4$$

Using the Far-Field Electric Field of Hertzian Dipole at the Origin … (1 of 2)

- Recall that the electric field of a Hertzian dipole of strength $I_0 l$ centered at the origin is given at a point $Q(R,\theta,\phi)$ in the far-field by

$$\tilde{E}_\theta = I_0 l \frac{jk_0\eta_0}{4\pi} \sin\theta \left(\frac{e^{-k_0 R}}{R}\right)$$

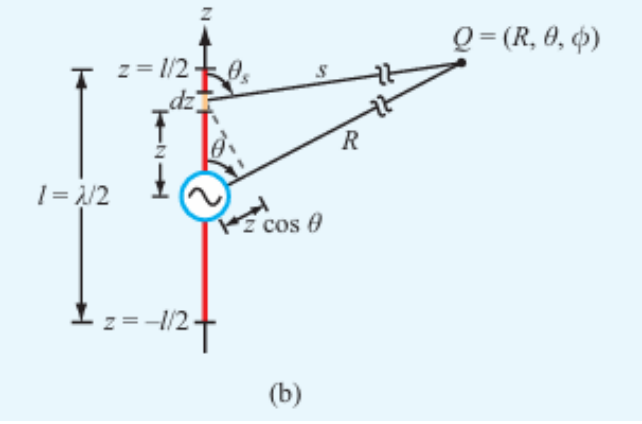
- From the above, we can deduce what is the contribution to the total electric field at a point $Q(R,\theta,\phi)$ in the far-field due to an infinitesimal length of the half-wave dipole centered at $(0,0,z)$

$$d\tilde{E}_\theta(z) = \tilde{I}(z)dz \frac{jk_0\eta_0}{4\pi} \sin\theta_s \left(\frac{e^{-k_0 s}}{s}\right)$$

Using the Far-Field Electric Field of Hertzian Dipole at the Origin …(2 of 2)

- Where s is the distance from the infinitesimal dipole to the point Q and θ_s is the angle from the z-axis to the distance vector from the infinitesimal dipole to the point Q.
- Since Q is very far away, we can use the ***parallel rays approximation:***

$$\theta_s = \theta$$
$$s = R - z\cos\theta\,,\ \text{for phase}$$
$$s = R,\ \text{for magnitude}$$



Far-Field Electric Field of Half-Wave Dipole

- Thus,

$$d\tilde{E}_\theta(z) = \tilde{I}(z)dz\frac{jk_0\eta_0}{4\pi}\sin\theta\left(\frac{e^{-k_0R}}{R}\right)e^{jk_0z\cos\theta}$$

- The electric field at Q due to the entire antenna can be computed as

$$\tilde{E}_\theta = \int_{-\frac{\lambda}{4}}^{\frac{\lambda}{4}} d\tilde{E}_\theta(z) = I_0\frac{jk_0\eta_0}{4\pi}\sin\theta\left(\frac{e^{-k_0R}}{R}\right)\int_{-\frac{\lambda}{4}}^{\frac{\lambda}{4}}\cos\left(\frac{2\pi z}{\lambda}\right)e^{j\frac{2\pi}{\lambda}z\cos\theta}dz$$

$$= j\frac{\eta_0}{2\pi}I_0\left(\frac{e^{-k_0R}}{R}\right)\left\{\frac{\cos\left[\frac{\pi}{2}\cos\theta\right]}{\sin\theta}\right\}$$

Time-Average Poynting Vector of Half-Wave Dipole

$$\vec{S}_{\mathrm{av}}(\vec{R}) = \hat{R}\,\frac{\left|\tilde{E}_\theta\right|^2}{2\eta_0}$$

$$= \hat{R}\left(\frac{\eta_0}{8\pi^2}\right) {I_0}^2\,\frac{1}{R^2}\left\{\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right\}^2$$

Radiation Intensity of Half-Wave Dipole

$$U(\theta,\varphi) = \lim_{R\to\infty} R^2\, S_{\mathrm{av}}(R,\theta,\varphi)$$

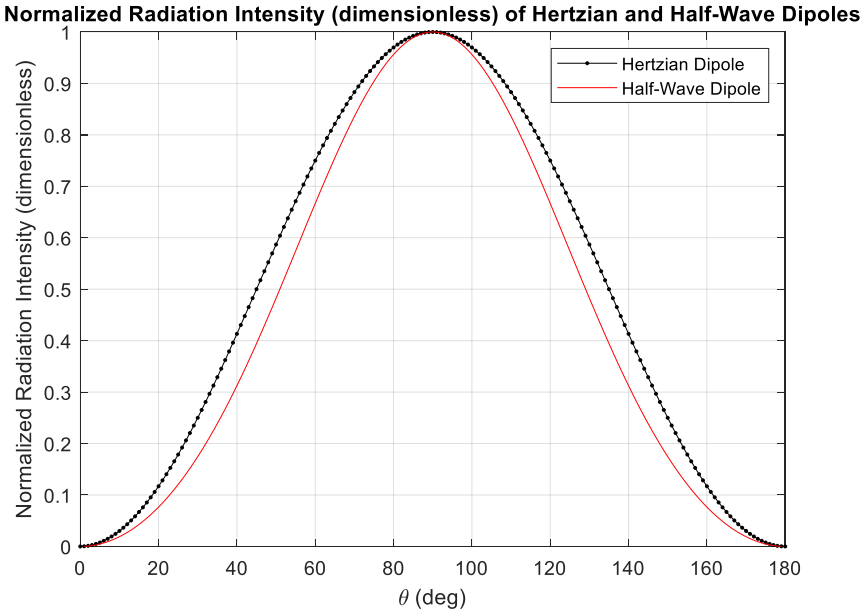
$$= \left(\frac{\eta_0}{8\pi^2}\right) {I_0}^2 \left\{\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right\}^2$$

Normalized Radiation Intensity of Half-Wave Dipole

$$F(\theta,\varphi) = \frac{U(\theta,\varphi)}{U_{\max}} \quad = \left\{\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\right\}^2$$

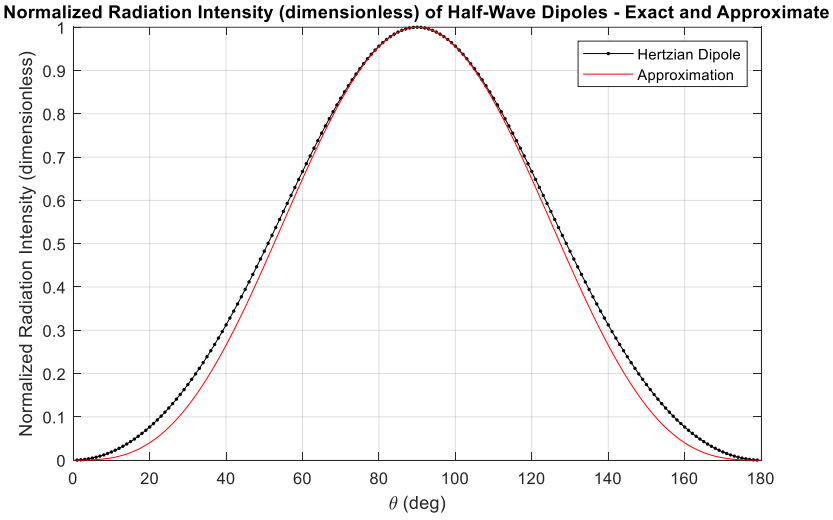
The half-wave dipole exhibits an omnidirectional radiation pattern

Normalized Radiation Intensity of Half-Wave Dipole



Approximation to Normalized Radiation Intensity of Half-Wave Dipole

$$F(\theta,\varphi) \approx \sin^3\theta$$



Question

- What is the directivity of the half-wave dipole? Compute the value using the approximation to the normalized radiation intensity.

Quarter-Wave Monopole

- A quarter-wave monopole placed over an infinite conducting plane produces the same radiation pattern in the upper half-space as a half-wave dipole, and zero radiation in the lower half-space.

$$F(\theta,\varphi) = \begin{cases} \left\{\dfrac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin\theta}\right\}^2 & ,0 \le \theta < 90^\circ \\ 0, & 90 \le \theta < 180^\circ \end{cases}$$

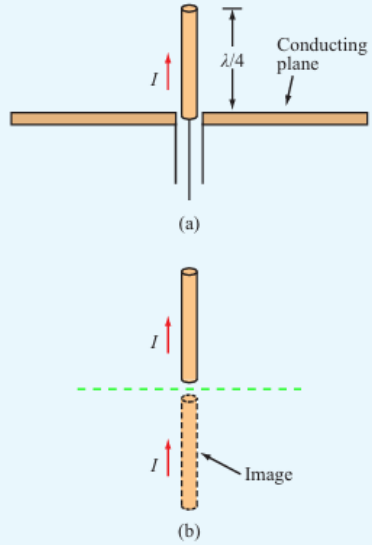


Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

Dipoles of Arbitrary Length

- Assume
  - The dipole is “thin” (radius of wire much less than wavelength and length of dipole).
  - The dipole is fed at its center.

Radiation Patterns for Dipoles of Arbitrary Length

- The radiation pattern of a straight wire of any length is omni-directional.
- For lengths less than one wavelength, the pattern has one lobe, and the directivity increases with length.
- For lengths greater than one wavelength, the pattern has multiple lobes.

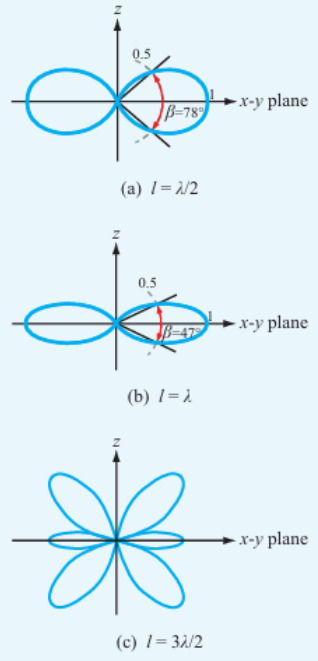


Figure 9-17 Radiation patterns of dipoles with lengths of $\lambda/2$, λ, and $3\lambda/2$.

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